

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA0706
NETWORKS

COURSE NAME: COMPUTER

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I Maniyam Anushka(192525411)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:Maniyam Anushka

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.
- ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also,

calculate the total number of possible host addresses available within this network based on the given prefix length.

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

SIMATS ENGINEERING

Assessment Tool 1 — Annexure A (Analytical Problem Solving)

Question 1 — CRC (Cyclic Redundancy Check)

Given:

Message = 11100011

Generator polynomial = $X^3 + 1 \rightarrow$ binary divisor = 1001

Formula:

Append 3 zeros $\rightarrow$ divide by 1001 using XOR $\rightarrow$ remainder = CRC bits

Calculation:

Dividend = 11100011 + 000 = 11100011000

$1110 \oplus 1001 = 0111 \rightarrow$ bring down 0 $\rightarrow 1110$

$1110 \oplus 1001 = 0111 \rightarrow$ bring down 0 $\rightarrow 1110$

$1110 \oplus 1001 = 0111 \rightarrow$ bring down 1 $\rightarrow 1111$

$1111 \oplus 1001 = 0110 \rightarrow$ bring down 1 $\rightarrow 1101$

$1101 \oplus 1001 = 0100 \rightarrow$ bring down 0 $\rightarrow 1000$

$1000 \oplus 1001 = 0001 \rightarrow$ bring down 0 $\rightarrow 0010$ (no XOR)

$0010 \rightarrow$ bring down 0 $\rightarrow 0100$ (no XOR)

Remainder (CRC) = 100

Transmitted Message = 11100011100

Part (ii) — Leftmost bit inverted:

Received message = 01100011100

Divide 01100011100 by 1001 $\rightarrow$ Remainder = 010

Receiver's Remainder = 010 ($\neq$ 000 $\rightarrow$ Error Detected)

Related:

- A remainder of 000 always means the frame is error-free.
 - A non-zero remainder tells the receiver the frame is damaged, so it is discarded and retransmission is requested.
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Question 2(a) — Throughput

Given:

Frames = 12,000 per minute

Each frame = 10,000 bits

Formula:

$$\text{Throughput} = (\text{Frames} \times \text{Bits}) / 60$$

Calculation:

$$\begin{aligned} &= (12000 \times 10000) / 60 \\ &= 2,000,000 \text{ bits/sec} \end{aligned}$$

$$\text{Throughput} = 2 \text{ Mbps}$$

Related:

- Bandwidth = Maximum capacity of the link
- Throughput = Actual data transferred (usually less than bandwidth)

Question 2(b) — Channel Capacity

Given:

Bandwidth (B) = 3000 Hz

Signal-to-Noise Ratio (SNR) = 3162

Formula:

$$C = B \times \log_2(1 + \text{SNR}) \quad (\text{Shannon's Theorem})$$

Calculation:

$$\begin{aligned} C &= 3000 \times \log_2(1 + 3162) \\ C &= 3000 \times \log_2(3163) \\ \log_2(3163) &\approx 11.63 \\ C &\approx 3000 \times 11.63 \approx 34,880 \text{ bits/sec} \end{aligned}$$

$$\text{Channel Capacity} \approx 34.88 \text{ kbps}$$

Related:

- Higher bandwidth or better SNR → higher channel capacity.
- This is the theoretical maximum reliable data rate for the channel.

Question 3 — Hamming (12,8) Code for Patient Monitoring

Scenario:

A hospital sends heart rate and blood pressure data wirelessly. Interference sometimes flips one bit. Since doctors need the data instantly, the hospital uses Hamming (12,8), which can correct a single-bit error immediately — no retransmission needed.

Encoding — how the 12 bits are built:

$$12 \text{ bits} = 8 \text{ data bits} + 4 \text{ parity bits}$$

- Parity bits sit at positions 1, 2, 4, 8 (powers of 2).
- Data bits fill positions 3, 5, 6, 7, 9, 10, 11, 12.
- P1 checks positions 1, 3, 5, 7, 9, 11
- P2 checks positions 2, 3, 6, 7, 10, 11
- P4 checks positions 4, 5, 6, 7, 12
- P8 checks positions 8, 9, 10, 11, 12
- Each parity bit is set so its group has an even number of 1's.

Detection & Correction at the receiver:

- Receiver redoes all 4 parity checks → gets a 4-bit syndrome.
- Syndrome = 0000 → no error.
- Syndrome $\neq$ 0000 → its value = exact position of the wrong bit (1–12).
- That one bit is flipped back — error corrected instantly.

*Single-bit errors are detected AND corrected instantly
— no retransmission delay for critical patient data*

Question 4(a) — IPv4 Subnetting

Given:

Network = 172.16.10.0/24 New mask = /27

Formula:

$$\text{Subnets} = 2^{(\text{borrowed bits})} \quad \text{Usable Hosts} = 2^{(\text{host bits})} - 2$$

Calculation:

$$\text{Borrowed bits} = 27 - 24 = 3$$

$$\rightarrow \text{Subnets} = 2^3 = 8$$

$$\text{Host bits} = 32 - 27 = 5 \rightarrow$$

$$\text{Usable hosts} = 2^5 - 2 = 30$$

$$\text{Block size} = 256 \div 8 = 32$$

The 8 subnets created:

Subnet Range	Subnet Range
172.16.10.0 – .31	172.16.10.32 – .63
172.16.10.64 – .95 ← contains .78 & .95	172.16.10.96 – .127
172.16.10.128 – .159	172.16.10.160 – .191
172.16.10.192 – .223	172.16.10.224 – .255

172.16.10.78 → Subnet 172.16.10.64/27
Network Address = 172.16.10.64
Broadcast Address = 172.16.10.95

Related:

- 172.16.10.95 IS inside this same subnet — it is the broadcast address, not a usable host.

Question 4(b) — IPv6 Addressing

Given:

Address = 2001:0db8:0000:0000:0000:ff00:0042:8329

Rules:

- Remove leading zeros in each group.
- Replace the longest run of all-zero groups with :: (only once).

Calculation:

Compressed → 2001:db8::ff00:42:8329
Expanded → 2001:0db8:0000:0000:0000:ff00:0042:8329
Network Prefix (/64) → 2001:db8:0000:0000::/64
Host bits = 128 – 64 = 64 → Host addresses = 2⁶⁴

Compressed = 2001:db8::ff00:42:8329
Total Host Addresses = 2⁶⁴

Question 5 — Online Examination Network Scenario

Problem Identified:

The server (10 Gbps) sends data much faster than many students' connections (5–100 Mbps) can receive. This is a Flow Control problem — a speed mismatch between sender and receiver.

Why it happens:

Each receiver has a limited buffer. When data arrives faster than the buffer can be emptied, the buffer overflows and extra packets are dropped.

Effects:

- Packet loss → incomplete downloads
- Retransmissions → extra network traffic
- Delayed acknowledgments → sender slows/times out
- Portal freezing → device can't process data fast enough

Fix:

Use flow control (receiver tells sender how much it can accept, e.g. sliding window) and congestion control (e.g. TCP slow start) so the sender's speed matches the receiver's capacity.

Root Cause: Flow Control Mismatch
(Sender transmission speed > Receiver processing speed)